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0178/01

May/June 2018

1 hour

Additional Materials: Geometrical Instruments
Tracing paper (optional)

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

If working is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.
The total of the marks for this paper is 60.



- 1 (a) Evaluate $6 + 27 \div 3$.

Answer (a) [1]

- (b) Find the value of $28 \div 3\frac{1}{4}$.

Answer (b) [1]

- (c) Work out $\frac{2}{3} - \frac{1}{4}$.

Answer (c) [2]

- 2 (a) (i) Express 130 as a product of its prime factors.

Answer (a)(i) [1]

- (ii) Find the smallest positive integer value n for which $130n$ is a multiple of 35.

Answer (a)(ii) [1]

- (b) Find the highest common factor (HCF) of 48 and 84.

Answer (b) [2]



- 3 (a) Express $\frac{9}{20}$ as a percentage.

Answer (a)% [1]

- (b) Write 0.28 as a fraction in its lowest terms.

Answer (b) [2]

- (c) Work out 0.1^3 .

Answer (c) [1]

- 4 Consider the given numbers.

0.4 $\frac{2}{3}$ $\sqrt{23}$ 1 2

Write down a number that is

- (a) prime,

Answer (a) [1]

- (b) irrational,

Answer (b) [1]

- (c) a square number.

Answer (c) [1]



- 5 The side of a square is given as 9 cm, correct to the nearest centimetre.

Find the least possible value of the side.

Answer cm [1]

- 6 (a) Express the ratio 400 cm to 1 km in its simplest form.

Answer (a) [2]

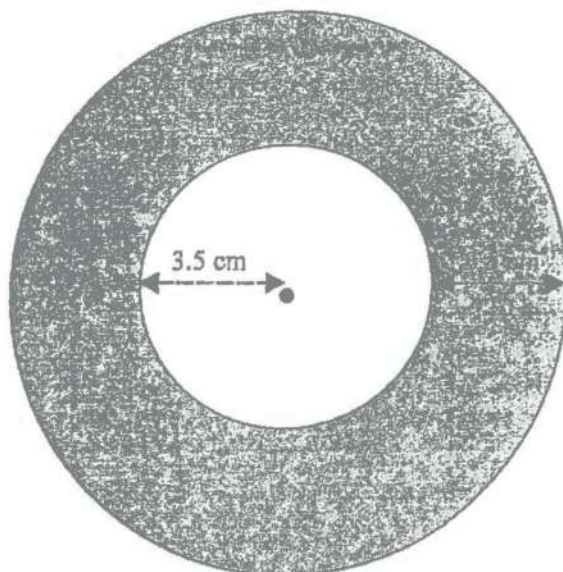
- (b) Write the numbers in order of size, starting with the **smallest**.

$$\frac{1}{2} \quad 0.2^2 \quad 0.25 \quad \sqrt{0.04}$$

Answer (b) < < < [2]



- 7 A paper ring is made by cutting round a circle of radius 7 cm and then cutting out a small circle of radius 3.5 cm as shown in the diagram.



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Use $\pi = \frac{22}{7}$.

Work out the

- (a) area of the ring,

Answer (a) cm^2 [3]

- (b) total perimeter of the ring.

Answer (b) cm [3]



8 Find the value of

(a) $15^4 \div 15^2$,

Answer (a) [1]

(b) $\left(\frac{1}{2}\right)^{-1}$,

Answer (b) [1]

(c) $125^{\frac{2}{3}}$.

Answer (c) [1]

9 (a) If $f(x) = 3x + 2$, find $f(-2)$.

Answer (a) [2]

(b) Factorise fully

$4x^2 - y^2$

Answer (b) [1]



- 10 The table shows part of a pattern of numbers and the sum of the numbers in each row.

Row number	Numbers	Sum
1	1	1
2	3,5	8
3	7,9,11	27
4	13,15,17,19	t
5		

- (a) Write down the value of t .

Answer (a) [1]

- (b) Complete row 5 in the table.

[2]

- (c) The table is continued to row n .

- (i) How many numbers would appear in row n ?

Answer (c)(i) [1]

- (ii) Write, in terms of n , an expression for the sum of numbers in row n .

Answer (c)(ii) [1]



- 11 A car travels 50 km at a speed of 40 km/h.

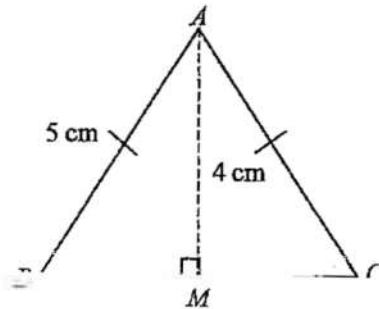
The journey starts at 12 10.

Work out the time the journey finishes.

Answer [3]

- 12 The diagram shows an isosceles triangle ABC .

M is a point on BC as shown.



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$AB = 5$ cm and $AM = 4$ cm.

Find

- (a) the length BC ,

Answer (a) cm [2]

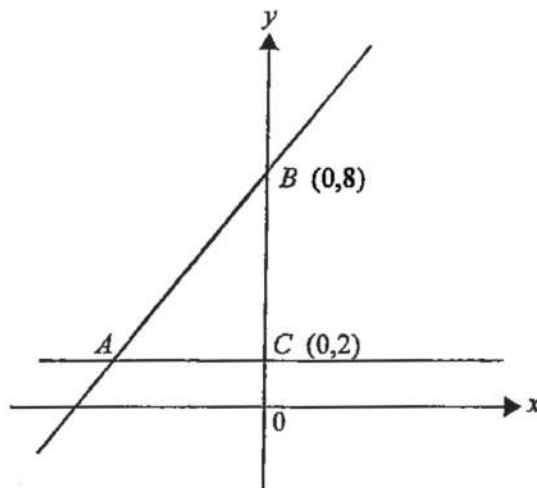
- (b) $\sin \hat{ACB}$.

Answer (b) [1]



- 13 In the diagram, B is the point $(0, 8)$ and C is the point $(0, 2)$.

The sloping line through B and the horizontal line through C meet at the point A .



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- (a) Write the equation of the line AC .

Answer (a) [1]

- (b) Given that the gradient of the line AB is 2, find the equation of the line AB .

Answer (b) [1]

- (c) Find the coordinates of A .

Answer (c) (..... ;) [2]

- (d) Work out the area of the triangle ABC .

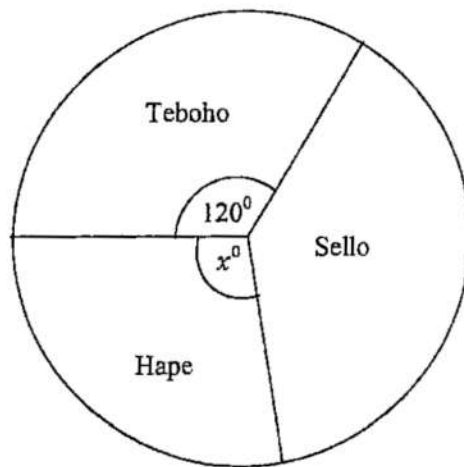
Answer (d) [2]



- 14 Teboho, Sello and Hape have 300 sweets to sell at school.

Hape sold 80 sweets.

The pie chart represents the number of sweets each boy sold.



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- (a) How many sweets were sold by Teboho?

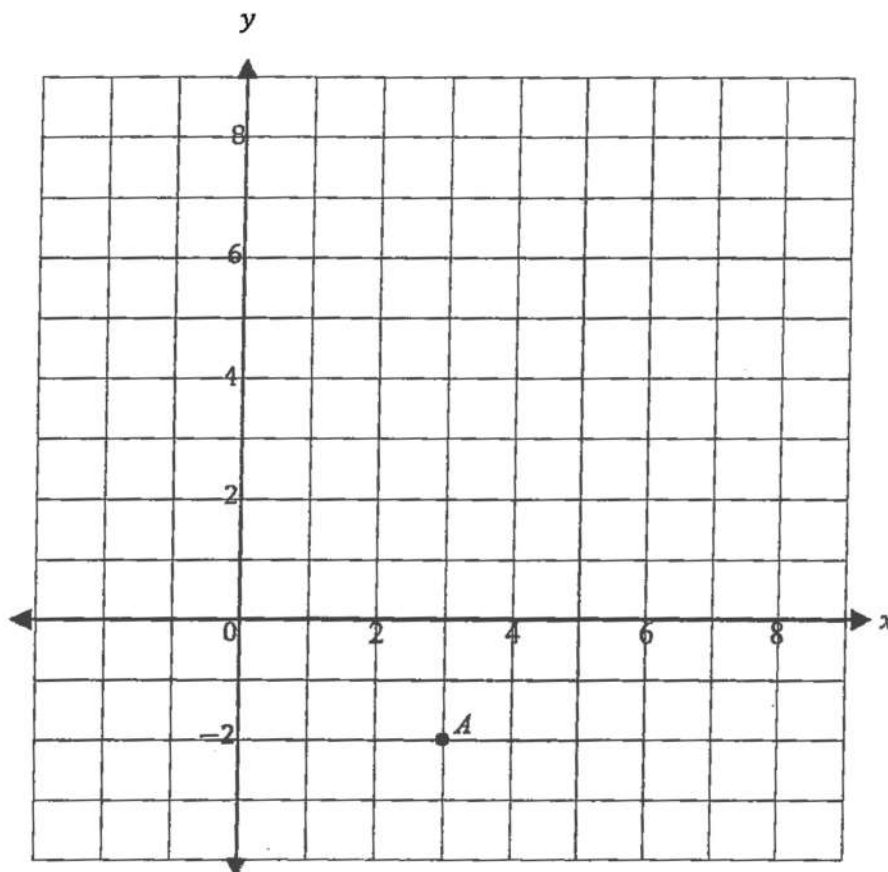
Answer (a) [2]

- (b) Find the value of x .

Answer (b) $x =$ ° [2]



- 15 In the diagram, A is the point $(3, -2)$.



Find the coordinates of the points B and C such that

- (a) B is the image of A after the reflection in the line $y = 2$,

Answer (a) (.....,) [2]

- (b) C is the image of A after translation by vector $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$.

Answer (b) (.....,) [2]



16 Four identical cards are numbered 1, 4, 6 and 9 respectively.

A card is chosen at random.

Find the probability that the card is numbered

(a) 9,

Answer (a) [1]

(b) even or odd,

Answer (b) [1]

(c) at least 7.

Answer (c) [1]



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